MAAT v1.2.1 Variable Closure and Measurement Definitions

Formal Definition of Structural Supports, Activity,
Response-Derived Closure, and Projection Observable Derivation

MAAT Structural Selection Series

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Abstract

This paper establishes the formal definition of all variables used in the MAAT v1.2.1 framework. It resolves remaining ambiguities in the measurement of the primary support sectors (H, B, S, V), the definition of activity, and the role of the closure coefficient λ_{cl}.

Three results are established. First, all support sectors are derived from non-negative defect measures with explicit physical definitions. Second, the closure coefficient λ_{cl} is derived from the primary response weights λ_a via a sensitivity projection — it introduces no additional primary sector. Third, the projection observable $\mathcal{C}_{\mathrm{proj}}(z)$, previously introduced phenomenologically, is connected to the response geometry of the structural defect ensemble through a canonical response map. Its three parameters (γ, B_*, α) are fixed once the defect covariance matrix, target defect vector, and response-map convention are specified.

The central result is an operationally closed formulation: all observable quantities are traceable either to defect measures or to response-based weights. Robustness is not an independent degree of freedom but a derived consistency bound on the joint support structure. No independent fifth sector is introduced.

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1 Introduction

The MAAT structural selection framework (Papers 01–40) has evolved from a diagnostic tool into a closed effective theory. The transition from v1.0 (five primary sectors including independent R) to v1.2.1 (four primary sectors with emergent robustness) was established in Papers 36–37.

A formal gap remained: the projection observable $\mathcal{C}_{\text{proj}}(z)$ and the closure coefficient λ_{cl} were introduced with phenomenological motivation but without explicit derivation from the primary defect structure.

This paper closes that gap at the effective-framework level. The result is a variable system in which every quantity — support scores, weights, robustness, projection parameters — is traceable to the defect covariance matrix C, the target defect vector d^*, and the explicitly stated response-map convention. No separately fitted projection parameter is introduced once these inputs are fixed.

2 Selection Functional

The MAAT v1.2.1 selection functional is:

$$\mathcal{F}_{\text{MAAT}}[\Phi] = -\sum_{a\in\{H,B,S,V\}} \lambda_a \log(\varepsilon + \Gamma_a[\Phi]) - \lambda_{\text{cl}} \log(\varepsilon + R_{\text{rob}}[\Phi]), \tag{1}$$

$$P[\Phi] \propto \exp(-\mathcal{F}_{\text{MAAT}}[\Phi])\,. \tag{2}$$

The key requirement: all terms must be derived from measurable or computable quantities. Table 1 summarises every symbol.

Table 1: Symbol reference for the MAAT v1.2.1 master equation.

Symbol	Range	Meaning
$d_a[\Phi]$	≥ 0	structural defect in sector a; zero = perfectly satisfied
Γ_a	$(0,1]$	support score $= 1/(1+d_a)$; high = structurally healthy
λ_a	≥ 0	response-derived selection pressure (Paper 30)
λ_{cl}	≥ 0	derived closure coefficient; not a primary sector weight
ε	$\ll 1$	regulariser preventing $\log(0)$
R_{resp}	$(0,1]$	implicit respect $= (HBV)^{1/3}$
R_{rob}	$(0,1]$	emergent robustness $= \min(R_{\text{resp}}, (HBSV)^{1/4})$
Γ_H	$(0,1]$	equation-of-motion consistency
Γ_B	$(0,1]$	energy-momentum conservation
Γ_S	$(0,1]$	controlled dynamical activity
Γ_V	$(0,1]$	connectedness / field coupling

3 Support Fields from Defect Measures

Each structural sector is defined via a non-negative defect $d_a \geq 0$:

$$\Gamma_a = \frac{1}{1 + d_a}, \qquad 0 < \Gamma_a \leq 1. \tag{3}$$

High Γ_a means low defect (structurally healthy); $\Gamma_a \to 0$ means critical defect. Table 2 gives the explicit defect definitions.

Table 2: Defect definitions for the four primary sectors.

Sector	Defect d_a	Physical meaning
H (Harmony)	$\|\mathcal{E}[\Phi]\|^2$	Squared norm of equation-of-motion residual; zero when field equations are satisfied exactly
B (Balance)	$\|\nabla_\mu T^{\mu\nu}\|^2$	Squared divergence of stress-energy tensor; zero when energy-momentum is conserved
S (Structure)	$\left(\frac{A[\Phi]-A_*}{\sigma_A}\right)^2$	Target-normalised activity defect; zero at controlled activity $A[\Phi] = A_*$, large for frozen or chaotic regimes
V (Connectedness)	$1 - \tilde{I}_{\mathrm{nn}}$	One minus normalised nearest-neighbour mutual information ($\tilde{I}_{\mathrm{nn}} \in [0, 1]$, see Section 3.4); high V = coherent coupling, low V = fragmentation

3.1 H — Harmony / Equation Consistency

$$d_H = \|\mathcal{E}[\Phi]\|^2, \tag{4}$$

where $\mathcal{E}[\Phi]$ is the residual of the equations of motion (e.g. $\mathcal{E}[\Phi] = \Box\Phi - V'(\Phi)$ for a scalar field). $H \approx 1$ indicates a dynamically consistent configuration; $H \ll 1$ indicates strong equation violation [7].

3.2 B — Balance / Constraint Consistency

$$d_B = \|\nabla_\mu T^{\mu\nu}\|^2. \tag{5}$$

This measures violation of energy-momentum conservation. In general relativity, $\nabla_\mu T^{\mu\nu} = 0$ is a Bianchi identity [8]; departures from it signal structural imbalance.

3.3 S — Creative Structure / Activity

Activity is defined relative to a target activity level A_* with dispersion σ_A:

$$\boxed{d_S = \left(\frac{A[\Phi] - A_*}{\sigma_A}\right)^2, \qquad A[\Phi] = \frac{1}{T}\int_0^T \left|\frac{\partial\Phi}{\partial t}\right|^2 dt.} \tag{6}$$

This formulation ensures that $S \approx 1$ when $A[\Phi] \approx A_*$ (controlled activity), not when $A[\Phi]$ is maximal. The sector thus selects the structurally active intermediate regime, not the most energetic one.

An information-theoretic proxy is available when time-series are not directly accessible:

$$d_S \sim \dot{S}_{\mathrm{cg}}^{(+)}, \tag{7}$$

the positive part of the coarse-grained entropy production rate [3]. Low S corresponds to a frozen system; high S to a chaotic or unstable one; $S \approx 1$ to controlled activity.

3.4 V — Connectedness

$$d_V = 1 - \tilde{I}_{\mathrm{nn}}, \qquad \tilde{I}_{\mathrm{nn}} = \frac{I_{\mathrm{nn}}}{\log|\mathcal{X}|} \in [0,1], \tag{8}$$

where I_{nn} is the nearest-neighbour mutual information between field values at adjacent spatial locations, and $\log|\mathcal{X}|$ is the maximum possible mutual information for the given alphabet [9]. The normalisation $\tilde{I}_{\mathrm{nn}} \in [0,1]$ ensures $d_V \geq 0$ always. Without normalisation, $I_{\mathrm{nn}} > 1$ is possible, which would give a negative defect and violate the requirement $d_a \geq 0$. High V indicates coherent spatial coupling; low V indicates fragmentation.

4 Emergent Robustness Closure

Closure equations (v1.2.1). Robustness is not an independent degree of freedom but a derived consistency bound on the joint support structure:

$$R_{\mathrm{resp}} = (H \cdot B \cdot V)^{1/3}, \tag{9}$$

$$R_{\mathrm{rob}} = \min\Big(R_{\mathrm{resp}},\ (H \cdot B \cdot S \cdot V)^{1/4}\Big). \tag{10}$$

R_{resp} is the minimal structural consistency (constraint adherence); R_{rob} is the full stability boundary (robustness under perturbation). Since $R_{\mathrm{rob}} \leq R_{\mathrm{resp}}$ always, robustness is strictly the weaker condition.

The structural stability diagnostic is:

$$\text{Stability} = R_{\mathrm{rob}}. \tag{11}$$

Why these geometric means? The geometric mean $(\cdot)^{1/n}$ is the unique symmetric, monotone, homogeneous-of-degree-one aggregation that corresponds to the maximum-entropy measure over sector defects subject to fixed total log-defect [1, 2]. The $\min(\cdot)$ structure implements the Onsager reciprocal principle: robustness is bounded by the weakest relevant sector [4].

5 Response-Derived Weights

The primary weights λ_a are obtained from the response closure of Paper 30 [10]:

$$\lambda = \left(C + \eta\,\frac{\mathrm{tr}(C)}{A}\,I\right)^{-1} (\langle d\rangle_{\mu_0} - d^*)\,, \tag{12}$$

where:

Symbol	Meaning
$C = \mathrm{Cov}_{\mu_0}(d_a, d_b)$	defect covariance matrix under reference measure μ_0
$\langle d \rangle_{\mu_0}$	mean defect vector under μ_0
d^*	target defect vector (desired structural state)
η	ridge regularisation parameter
A	number of sectors (here $A = 4$)

The raw response weights λ_a^{raw} may take any sign. In Eq. (1), λ_a denotes the non-negative penalty weights $\lambda_a^{(+)}$, while λ_a^{raw} are the unconstrained response weights obtained from Eq. (12):

$$\lambda_a^{(+)} = \max(0,\, \lambda_a^{\mathrm{raw}})\,. \tag{13}$$

For the normalised response shares used in the projection parameter derivation (Section 8), the absolute values are used to capture the full response geometry including negative pressures:

$$\pi_a = \frac{|\lambda_a^{\mathrm{raw}}|}{\sum_b |\lambda_b^{\mathrm{raw}}| + \varepsilon}. \tag{14}$$

This distinction is relevant when extended response formulations allow negative raw weights. This follows the Onsager–Kubo response-theory tradition [4, 5].

6 Closure Coefficient λ_{cl}

The closure coefficient weights the derived robustness term in Eq. (1). It must be derived from the primary weights λ_a without introducing new degrees of freedom.

Definition.

$$\lambda_{\mathrm{cl}} = \sum_{a \in \{H,B,S,V\}} w_a\, \lambda_a, \tag{15}$$

with sensitivity weights

$$w_a = \frac{|\partial R_{\mathrm{rob}}/\partial \Gamma_a|}{\sum_b |\partial R_{\mathrm{rob}}/\partial \Gamma_b|}. \tag{16}$$

6.1 Interpretation

w_a measures how strongly the robustness boundary R_{rob} responds to a unit change in sector a. The closure coefficient λ_{cl} therefore projects the primary selection pressures onto the robustness boundary with the appropriate sensitivity weights.

- λ_{cl} is not a fifth sector weight.
- It introduces no additional degrees of freedom.
- It is fully determined by λ_a and the local geometry of R_{rob}.

6.2 Analytical gradients

The gradients of R_{rob} with respect to the support scores can be computed analytically. In the regime where $R_{\mathrm{resp}} = (HBV)^{1/3}$ is the active minimum:

$$\frac{\partial R_{\mathrm{rob}}}{\partial H} = \tfrac{1}{3}(HBV)^{-2/3} BV, \qquad \frac{\partial R_{\mathrm{rob}}}{\partial B} = \tfrac{1}{3}(HBV)^{-2/3} HV,$$
$$\frac{\partial R_{\mathrm{rob}}}{\partial V} = \tfrac{1}{3}(HBV)^{-2/3} HB, \qquad \frac{\partial R_{\mathrm{rob}}}{\partial S} = 0. \tag{17}$$

In the regime where $(HBSV)^{1/4}$ is active:

$$\frac{\partial R_{\text{rob}}}{\partial a} = \frac{1}{4}(HBSV)^{-3/4} \prod_{b \neq a} \Gamma_b, \quad a \in \{H, B, S, V\}. \tag{18}$$

The boundary switches at $R_{\text{resp}} = (HBSV)^{1/4}$, i.e. when $(HBV)^{1/3} = (HBSV)^{1/4}$.

7 Derivation of the Projection Observable

7.1 Structural breadth and coherent depth

Two fundamental contributions define the projection layer.

Structural breadth.

$$B_{\text{exp}}(z) = E(z)(1+z), \tag{19}$$

where $E(z) = H(z)/H_0$ is the dimensionless Hubble expansion rate. This measures how strongly structures are expressed in the observable domain.

Coherent depth.

$$D_{\text{acc}}(z) = D(z)\, L(z), \tag{20}$$

where $D(z)$ is the linear growth factor [6] and $L(z)$ is a bounded latent-depth function. This measures how much structural information can remain coherently organised.

7.2 Bounded structural activation

Structural breadth cannot grow without bound. The saturating activation:

$$S_{\text{proj}}(z) = B_{\text{exp}}(z)\, \tanh\left[\left(\frac{B_{\text{exp}}(z)}{B_*}\right)^{\alpha}\right], \tag{21}$$

ensures linear response at small B_{exp}, saturation at large B_{exp}, and bounded projection behaviour.

7.3 Projection observable

$$\boxed{\mathcal{C}_{\text{proj}}(z) = \frac{S_{\text{proj}}(z)}{1 + \gamma\, D_{\text{acc}}(z)},} \tag{22}$$

The numerator describes how strongly structure is expressed; the denominator describes how strongly coherence suppresses this expression. Projection arises when structural breadth exceeds the coherent depth that can stabilise it.

7.4 Projection imbalance

$$R_{\text{proj}}(z) = \frac{|S_{\text{proj}}(z) - \gamma D_{\text{acc}}(z)|}{S_{\text{proj}}(z) + \gamma D_{\text{acc}}(z) + \varepsilon}. \tag{23}$$

This bounded quantity measures the mismatch between expression and coherence, and provides the transition marker z_{tr} studied in Papers 33–40.

8 Response-Derived Projection Map

The projection observable Eq. (22) contains three phenomenological parameters: γ, B_*, α. In MAAT v1.2.1 these are fixed by a canonical response map from the response weights λ_a.

8.1 Normalised response shares

The normalised shares π_a are defined using absolute raw response weights:

$$\pi_a = \frac{|\lambda_a^{\text{raw}}|}{\sum_{b\in\{H,B,S,V\}} |\lambda_b^{\text{raw}}| + \varepsilon}. \tag{24}$$

8.2 Parameter derivation

Response-derived projection parameters.

$$\gamma_\lambda = \frac{\pi_V + \pi_B}{\pi_S + \varepsilon}, \tag{25}$$

$$B_{*,\lambda} = \frac{1}{\pi_H + \pi_B + \varepsilon}, \tag{26}$$

$$\alpha_\lambda = 1 + \frac{\pi_S}{\pi_V + \varepsilon}. \tag{27}$$

8.3 Physical interpretation

These forms define the canonical response map used in Papers 33–40. They are chosen to respect dimensional consistency, positivity, and sectoral competition: stabilising sectors (B, V) enter the denominator of γ_λ, while the activity sector S sets the effective projection scale via $B_{*,\lambda}$ and α_λ. The formulas are therefore not fitted to the growth residuals; they encode how structural breadth, coherence, and activity compete in the response convention adopted here.

Parameter	Structural interpretation
$\gamma_\lambda = (\pi_V + \pi_B)/\pi_S$	Coherent-depth damping: connectedness and balance determine how strongly projected structure is stabilised
$B_{*,\lambda} = 1/(\pi_H + \pi_B)$	Activation scale: equation consistency and conservation determine how much breadth can be expressed before saturation
$\alpha_\lambda = 1 + \pi_S/\pi_V$	Activation sharpness: structural activity relative to connectedness determines how abruptly projection turns on

8.4 Fully derived projection

The response-derived projection observable is:

$$\boxed{C_{\text{proj},\lambda}(z) = \frac{S_{\text{proj},\lambda}(z)}{1 + \gamma_\lambda D_{\text{acc}}(z)}, \qquad C_{\text{proj},\lambda}(z) = C_{\text{proj}}(z;\, \lambda_H, \lambda_B, \lambda_S, \lambda_V).} \tag{28}$$

In this convention the projection observable is no longer an independently fitted template. It is determined by the response geometry of the structural defect ensemble together with the explicit response map above.

9 CCI Diagnostics

Two CCI forms are used in Papers 33–40.

Minimal CCI (primary fields only):

$$\mathrm{CCI}_{\mathrm{min}} = \frac{S}{H + B + V + \varepsilon}. \tag{29}$$

Robustness-weighted CCI:

$$\mathrm{CCI}_{\mathrm{diag}} = \frac{S}{H + B + V + R_{\mathrm{rob}} + \varepsilon}. \tag{30}$$

Important: $R_{\mathrm{resp}} = (HBV)^{1/3}$ must *not* be added to the denominator separately. It is already encoded in H, B, V. Adding it would double-count the same information and violate the independence requirement for the selection functional [2].

This separation ensures that the diagnostic CCI remains a reduced observable and not a reparameterisation of the full selection functional $\mathcal{F}_{\mathrm{MAAT}}$. CCI measures the ratio of activity to coherence; $\mathcal{F}_{\mathrm{MAAT}}$ measures the total selection cost. They are complementary, not equivalent.

CCI form	Denominator	Use case
$\mathrm{CCI}_{\mathrm{min}}$	$H + B + V$	Minimal, theory-level
$\mathrm{CCI}_{\mathrm{diag}}$	$H + B + V + R_{\mathrm{rob}}$	Diagnostic, includes stability margin

10 Closure Summary

MAAT v1.2.1 is operationally closed.

Quantity	Source	Free?
$\Gamma_a = 1/(1 + d_a)$	defect measure	no
λ_a	response closure Eq. (12)	no
$R_{\mathrm{resp}} = (HBV)^{1/3}$	Eq. (9)	no
$R_{\mathrm{rob}} = \min(R_{\mathrm{resp}}, (HBSV)^{1/4})$	Eq. (10)	no
$\lambda_{\mathrm{cl}} = \sum_a w_a \lambda_a$	Eq. (15)	no
$\gamma_\lambda, B_{*,\lambda}, \alpha_\lambda$	Eqs. (25)–(27)	no
$C_{\mathrm{proj},\lambda}(z)$	derived from above	no
Input: defect covariance matrix C and target d^*		
Output: all observable quantities		

No independent fifth sector. No separately fitted projection parameters once the response map is chosen. No fitted closure coefficient. The framework is traceable to the defect covariance geometry and the stated closure conventions.

The framework makes its phenomenological choices explicit: all effective parameters are fixed by the covariance geometry of the defect ensemble after the response-map convention has been specified.

The full derivation pipeline is:

$$d \;\to\; C = \mathrm{Cov}(d_a, d_b) \;\to\; \lambda_a \;\to\; \pi_a \;\to\; \gamma_\lambda,\, B_{*,\lambda},\, \alpha_\lambda \;\to\; C_{\mathrm{proj},\lambda}(z).$$

11 Status, Limitations, and Reproducibility

This paper is a formal closure and measurement-definition note. It does not introduce a new numerical data set, likelihood analysis, or observational fit. Its role is to make the variables used in Papers 33–40 explicit and reproducible.

- The defect definitions are effective diagnostics. They are intended to be computable in field, cosmological, and constraint-system benchmarks, but they are not claimed to be unique microscopic observables.
- The projection map $(\lambda_H, \lambda_B, \lambda_S, \lambda_V) \mapsto (\gamma_\lambda, B_{*,\lambda}, \alpha_\lambda)$ is the canonical convention used in this series. Alternative response maps can be tested as ablations.
- No new external observational data are used in this paper. Data attribution and licence notes for the chronometer, Planck-normalised, and growth-data benchmarks remain those stated in Papers 28–40 and in the repository experiment READMEs.

The associated repository is:

https://github.com/Chris4081/structural-selection-principle

The relevant reproducibility folders for the variables defined here are:

Folder	Role
`experiments/lambda_response_closure/`	Response-based weight closure used for λ_a and λ_{cl}.
`experiments/maat_cci_projection_paper33/`	Projection observable and sensitivity tests.
`experiments/maat_paper39_observable_growth_signature/`	Observable growth-signature proxy using v1.2.1 closure.
`experiments/maat_paper40_structural_signature_test/`	Residual-structure signature test using $\mathrm{CCI}_{\mathrm{diag}}$.

12 Conclusion

Paper 41 establishes the formal closure of the MAAT v1.2.1 variable system. All quantities used in Papers 33–40 are now:

1. **Mathematically defined**: each sector has an explicit defect formula traceable to field quantities.
2. **Non-redundant**: no variable is added that is already encoded by others (in particular, R_{resp} is not double-counted in CCI).
3. **Free of hidden parameters**: the three projection parameters (γ, B_*, α) are fixed by the stated response map; the closure coefficient λ_{cl} is derived from primary weights and robustness sensitivity; no external fitting is required after the response convention is fixed.

This establishes MAAT v1.2.1 as a structurally closed effective framework, ready for:

- parameter-free structural signature tests (Paper 40),
- perturbation-level cosmological modelling (next paper),
- comparison with observational likelihood pipelines.

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